Chicago Traffic Crash Risk Scoring

Route Safety Intelligence via CRISP-DM

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Agenda

1. **Business Understanding**
   Problem, stakeholders, goals
2. **Data Understanding**
   Dataset, quality, spatial & temporal patterns
3. **Data Preparation**
   Cleaning, severity weights, feature engineering
4. **Modeling**
   Approach, architecture, validation design
5. **Evaluation**
   Metrics, feature importance, scenario multipliers
6. **Recommendations**
   What to deploy, trade-offs for stakeholders
7. **Conclusion**
   Key findings, limitations, decision takeaway
8. **Future Work**
   Next experiments, data needs, roadmap

The Problem

Core Question

"Given current or selected conditions, which Chicago zones are relatively more dangerous than usual?"

Scale of the problem:

- **USA:** ~6–7M crashes/year; 40,901 fatalities in 2023 (NHTSA)
- **Economic cost:** $340B–$1.4T annually; avg. $50K per crash (FHWA)
- **Illinois 2024:** 303,913 crashes; 1,085 fatal; $15B annual burden
- Chicago alone: 1,045,043 records since 2013 - raw point cloud is unactionable

Stakeholders

- Drivers, cyclists, pedestrians
- City agencies (Vision Zero)
- Logistics & freight operators
- Insurers & navigation platforms

Decision to Support

Rank Chicago grid cells 0–100 under any user-selected scenario so stakeholders can choose safer routes or prioritise infrastructure investment.

Project Goals & Success Criteria

Analytical goals:

- Zone-level risk ranking (relative, not absolute probability)
- Severity-weighted scoring - not just crash frequency
- Condition-switchable layers (rain / dark / wet)
- Temporal generalisation: train 2018–2024, validate 2025

Out of scope:

- Trip-level crash probability
- Causal inference
- Exposure-adjusted safety rates

Three-level Success Criteria

Business: clear zone tiers; switchable layers; interpretable explanations

Analytical: outperform naive hotspot baseline; stable top-K recovery; zero data leakage

MVP: grid CSV ready for map frontend; on-demand scoring API; daily refresh pipeline

Dataset Overview

Three linked CPD tables

Table	Rows	Cols
Crashes	1,045,043	48
Vehicles	2,130,727	71
People	2,293,745	29

Reliable modeling window: 2018–2025 (817,947 rows; pre-2018 coverage insufficient)

Key condition distributions:

- Weather: 78% CLEAR, 8% RAIN, 6% UNKNOWN
- Surface: 73% DRY, 13% WET, 10% UNKNOWN
- Lighting: 64% DAYLIGHT, 22% DARKNESS-LIGHTED

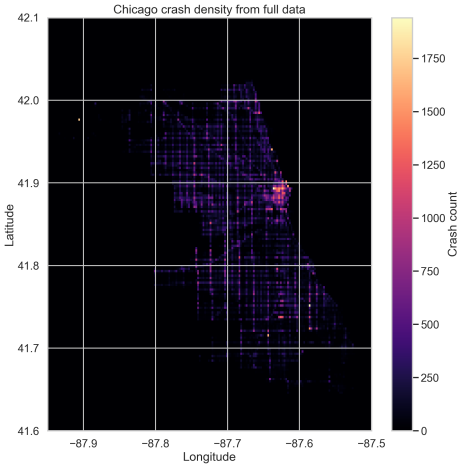
Key quality findings:

- Missing geo-coordinates: only 0.77%
- 4–10% UNKNOWN labels in condition fields - retained as explicit category
- No traffic volume denominator available - results must be relative risk

Main Limitation

Without trip exposure data we cannot compute risk per journey - only risk per zone per period. All scores are comparative, not absolute.

Geographic Crash Distribution

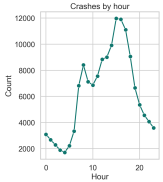


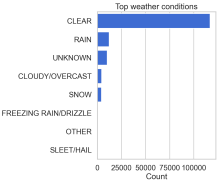
What this shows:

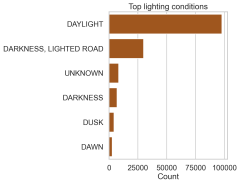
- Clear spatial clustering along arterial corridors
- Dense hotspots on North Side and downtown
- Residential zones have substantially lower raw counts

Business implication:
Raw density predicts future hotspots, but volume alone does not capture *severity* - a low-frequency zone with frequent injuries may deserve higher priority than a high-count minor-collision corridor.

Temporal & Condition Patterns





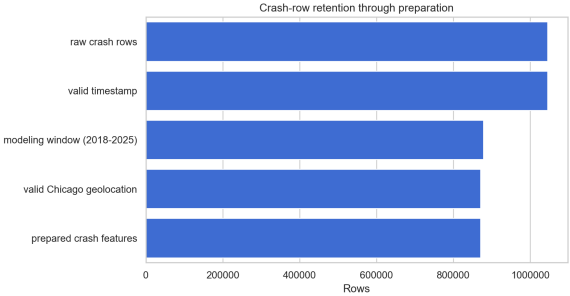


Key patterns:

- Morning and evening rush hours show pronounced peaks
- Rain multiplies crash rates even though clear conditions dominate by volume
- Darkness - even with street lighting - correlates with higher severity outcomes

Business implication:
Condition-aware scoring is essential. A static heatmap misses periods when the same zone becomes far more dangerous.

Cleaning & Filtering



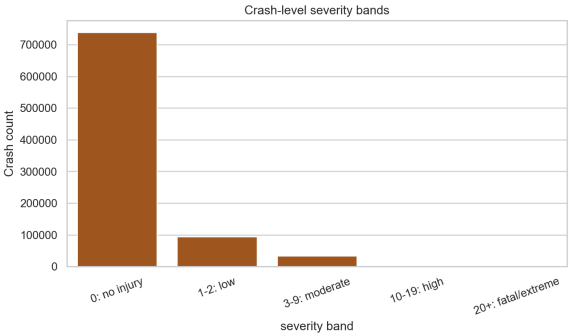
Steps taken:

1. Removed invalid timestamps and out-of-bounds coordinates
2. Normalised categorical fields; encoded UNKNOWN explicitly
3. Excluded post-crash leakage fields from feature set
4. Restricted to reliable window 2018–2025

Why it matters:
Including pre-2018 sparse data or leakage fields would inflate apparent model accuracy while producing unreliable real-world scores.

Final prepared dataset: **870,705 rows**.

Severity Weighting Design

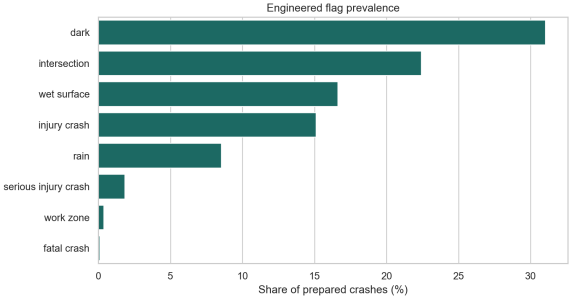


Why weight by severity?
Raw crash counts treat a fender-bender equally with a fatal collision. Severity weights align scores with real-world harm:

Outcome	Weight
Reported, not evident	1
Non-incapacitating	2
Incapacitating	5
Fatal	20

Business implication:
Zones with rare but serious injuries rank higher than zones with many low-severity events - matching stakeholder priorities around injury prevention.

Feature Engineering

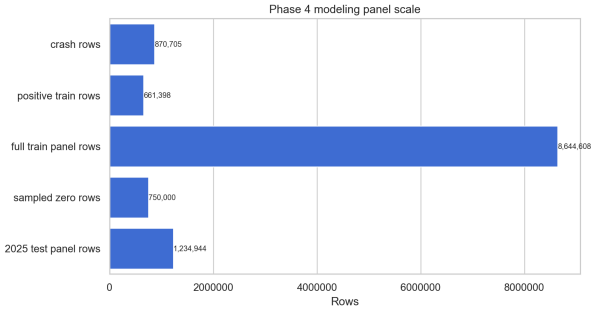


Feature groups built (4 groups):

- **Temporal:** hour, day-of-week, rush hour, night, season, sine/cosine encodings
- **Spatial:** ∼500 m grid cell ID, centroid lat/lon, historical priors (per-grid mean severity, slot mean)
- **Road context:** speed limit, traffic control, trafficway type, intersection flag, work zone
- **Scenario flags:** `is_rain`, `is_dark`, `is_wet_surface`, `is_pedbike`

Total: **48 model features** across 4 groups.

Modeling Approach



Two-layer design:

1. **ML model** - Histogram Gradient Boosting Regressor predicts `log1p(severity_weight_sum)` per grid-time slot
2. **Scenario multipliers** - smoothed condition adjustments applied at query time (no retraining needed)

Why Gradient Boosting?

- Handles mixed numeric/categorical natively
- Robust to sparse grid-time slots
- Fast inference for on-demand scoring
- Interpretable via permutation importance

Model Configuration & Validation Design

Phase 4 model settings

Target	severity_weight_sum on grid_id + crash_year + crash_month + crash_hour + is_weekend
Model	HistGradientBoostingRegressor trained on log1p(severity_weight_sum)
Train/test split	2018-2024 train, 2025 holdout test
Zero-row sampling	full train panel 8,644,608; positives 661,398; sampled zeros 750,000
Weights	positive rows 1.0; sampled zero rows 10.64428
Scenario layer	smoothed grid severity-per-crash multipliers, clipped to stable bounds

Temporal split (anti-leakage):

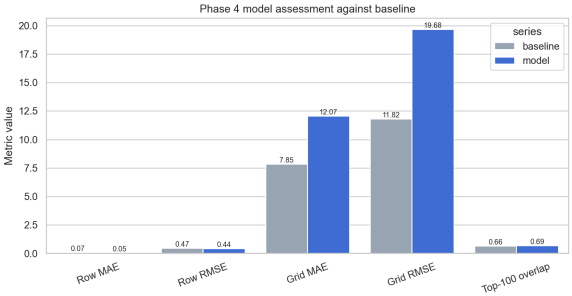
- **Train:** 2018–2024
- **Holdout test:** 2025 (full calendar year)

Grid-time panel construction:

- 8,644,608 total grid-time slots
- 661,398 slots with crashes (8%)
- Zero-slots explicitly included to teach the model what "safe periods" look like
- Zero-slot sampling (750k) with inverse frequency weighting

Key design choice: Zero-filling prevents the model learning only crash-present patterns - critical for reliable low-risk zone identification.

Model Performance vs. Baseline



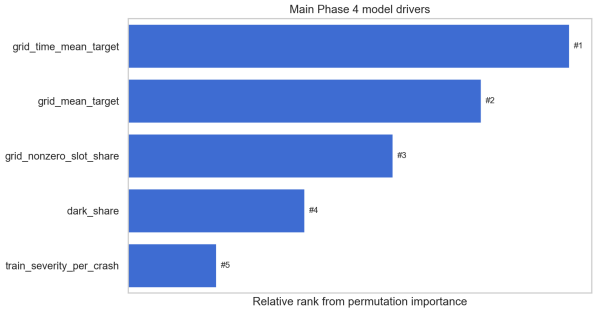
2025 holdout results:

Metric	Baseline	Model
MAE	0.072	**0.054**
RMSE	0.473	**0.444**
R^2	—	**0.35**
Top-100 overlap	0.66	**0.69**

Business interpretation:

- 26% MAE improvement - individual slot predictions meaningfully more accurate
- Top-100 overlap 0.69 means **69 of 100 high-risk zones** correctly predicted
- Zone ranking stable enough for operational use

What Drives Risk - Feature Importance



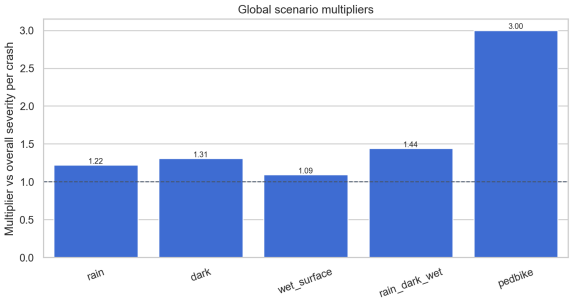
Top drivers:

1. Historical grid-time mean severity
2. Overall grid mean severity
3. Non-zero slot share (crash frequency)
4. Darkness share of the grid
5. Severity-per-crash ratio

Business interpretation:

- History dominates - past dangerous zones remain dangerous; no noise overfitting
- Darkness is a *structural* zone property - zones with poor lighting score higher even in clear conditions
- Severity-per-crash reveals injury-prone zones hidden by low raw crash volume

Condition Scenario Multipliers



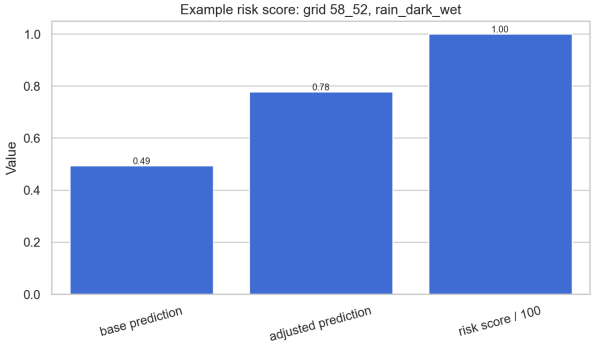
Empirically derived multipliers:

Scenario	Multiplier
Rain only	$\times 1.22$
Darkness only	$\times 1.31$
Wet surface only	$\times 1.09$
Rain + Dark + Wet	$\times 1.44$
Pedestrian / Bike	$\times 3.00$

Business interpretation:

- Combined rain-dark-wet lifts expected severity by **44%**
- Pedestrian/cyclist exposure is **3**$\times$ baseline - critical for active-transport routing
- Scenario switching requires no model retraining - instant response to changing conditions

Example Risk Score Output



Worked example:
Grid 58_52, weekday 18:00, November
Scenario: *rain_dark_wet*

Step	Value
Base ML prediction	0.494
After scenario adj.	0.777
Final 0–100 score	**100**

Business interpretation:

- Score 100 = top-tier risk; avoid or use maximum caution
- Same grid at midday, clear conditions scores substantially lower - conditions matter
- Output is a simple lookup: frontend needs no ML infrastructure

Recommendations for Stakeholders

What to deploy now

- Use precomputed grid scores as backend for interactive Chicago safety map
- Enable six scenario layers: overall, rain, dark, wet surface, combined, ped/bike
- Surface top-5 risk drivers per zone so users understand *why* a zone scores high
- Prioritise ped/bike mode for cycling-route and pedestrian-safety applications

Trade-offs to communicate

- Scores are **relative**, not probabilities - score of 80 $\neq$ 80% crash chance
- Historical priors dominate - new roads or infrastructure changes take time to be reflected
- $\sim$500 m resolution may conflate safe and dangerous micro-locations

Priority action

Connect a live weather API to auto-select scenarios so the map reflects current conditions without manual user input.

Conclusion

All success criteria met:

- ✓ 2,144 grid cells ranked 0–100
- ✓ Six deployable scenario layers
- ✓ 26% MAE improvement over naive baseline
- ✓ 0.69 top-100 hotspot overlap on unseen 2025 data
- ✓ Strict temporal split - zero leakage
- ✓ Flat-file outputs ready for frontend
- ✓ Interpretable feature drivers

Known limitations:

- No traffic-volume denominator - relative risk only
- 4–10% UNKNOWN condition labels dilute scenario signal
- Fixed grid resolution may hide within-cell variation
- No confidence intervals on scores

Decision takeaway

Model is production-ready as a relative risk-ranking backend.

Deploy for: route safety guidance, infrastructure prioritisation (Vision Zero), condition-aware alerts.

Treat scores as **comparative tiers**, not absolute probabilities.

Future Work

Near-term improvements:

- Incorporate vehicle and people context features (prepared but not yet in model)
- Rolling cross-validation within training window
- Systematic hyperparameter search
- Quantile regression for prediction intervals

Data enhancements:

- Live weather API for auto scenario selection
- Traffic exposure proxies (AADT, transit ridership)
- Road-segment-level granularity
- Real-time traffic feeds

Operational roadmap (Phase 6):

1. Build interactive map frontend
2. Automate data refresh and monthly retraining
3. Expose lightweight scoring API
4. Add crash-trend alerting for zones showing increasing severity
5. Causal policy evaluation - measure impact of infrastructure changes on scores

Confidence-building priority

Adding prediction intervals and exposure denominators would allow stakeholders to act on scores with higher statistical confidence and support formal policy decisions.

Thank You

Chicago Traffic Crash Risk Scoring

All six CRISP-DM phases complete. Project ready for deployment.

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